Lexical and morphosyntactic predictions are different: MEG studies in Hindi and Nepali

Introduction. Prediction of upcoming material is a key component of language comprehension. What do we predict exactly, and when? Neural responses evoked by unexpected material show that comprehenders predict semantic classes of words [1–4], specific lexical items [5], syntactic category [6], and agreement features [7]. But, less is understood about how, when and where the brain encodes these predictions. Some theories of agreement processing crucially rely on anticipatory encoding of inflectional feature [7–12], and some findings show neural activity encodes features of a word prior to encountering it [5, 13]. Here, we present two studies that leverage the split-ergative case systems of Hindi and Nepali to answer this question. We find that comprehenders anticipatorily encode morphosyntactic features of the verb (tense/aspect, agreement) and lexico-semantic features of the verb at distinct points in time and space. Thus, lexical prediction is not a unitary process in the brain.

Hindi and Nepali are subject-object-verb (SOV) languages, in which the subject and object's case-marking affect the expected inflectional features of the verb, including agreement (gender, person, number) and tense/aspect features [14, 15], both of which affect EEG responses at unexpected verbs in Hindi readers [16, 17]. In **Experiment 1,** we show in a sentence-completion task that simple subject+object prompts in Hindi and Nepali elicit a range of verb stem, verb gender, and verb tense/aspect completions that are partially independent, and vary with case marking. In **Experiment 2,** we show that MEG signals before the verb are sensitive to the entropy over these variables in different regions.

Methods. EXPERIMENT 1. Participants (N = 76 Hindi, N = 48 Nepali) completed 50 simple subject+object sentence prompts. Subject case (ergative, nominative), object case (bare, accusative), and object number (SG, PL) were manipulated in a 2×2×2 Latin-Square design (object number not reported; Fig 1A/B). EXPERIMENT 2. Participants (N = 22 Hindi, N = 22 Nepali) read sentences phrase-by-phrase while MEG signals were recorded. 25% of trials contained a picture-matching task for comprehension. Subject case, object case, and verb cloze (Low vs. High Cloze) were manipulated in a 2×2×2 within-subjects design, with each verb presented in the simple perfective after ergative subject NPs and past imperfective for nominative subject NPs (Fig 1A/D). Results of factorial design reported in [18].

Results. EXPERIMENT 1. Completions were coded for the verb stem, gender, and tense/aspect (Fig 1B). In Hindi, entropy, or negative weighted log sum of the response ($H(\mathbf{X}) = -\Sigma\, p(x) \log_2 p(x)$) over verb completions was correlated with verb root, verb gender, and verb aspect (ps < 0.01), and verb gender was correlated with verb aspect ($p < 0.01$). In Nepali, entropy over completions was correlated with verb root and verb aspect (ps < 0.01), and verb root was correlated with verb aspect ($p < 0.01$) (Fig 1C). EXPERIMENT 2. MEG data recorded during the 1000ms epoch corresponding to the pre-verbal object NP were projected to source space. MEG activity was averaged in coordinates in targeted regions of interest (ROIs), selected based on prior studies [18, 19] (Fig 1E). Log-likelihood ratio tests were conducted on each time point in each ROI, comparing a full model consisting of the entropy values over the coded features from Experiment 1 vs. a reduced model that removed one key factor (Fig 1F). Adjacent time points greater than $\chi^2(1) = 3.841$ ($p < 0.05$) were clustered together. Cluster-level p-values were estimated using temporal cluster-based permutation tests [20]. Significant clusters were observed for verb aspect in LIFG, verb root in LATL, verb completion in LPTL, and root and gender in LTPJ (Fig 1G).

Conclusion. In Experiment 1, we showed that in Hindi and Nepali, the predictability of tense/aspect, inflection, and verb stem were partially independent. In Experiment 2, we showed that syntactic and morphosyntactic information are encoded in LIFG and LTPJ, compatible with theories that attribute (morpho)syntactic processes to these areas [18, 21], and lexico-semantic information is encoded in LATL, LPTL, and LTPJ, key nodes in lexico-semantic processing [22].

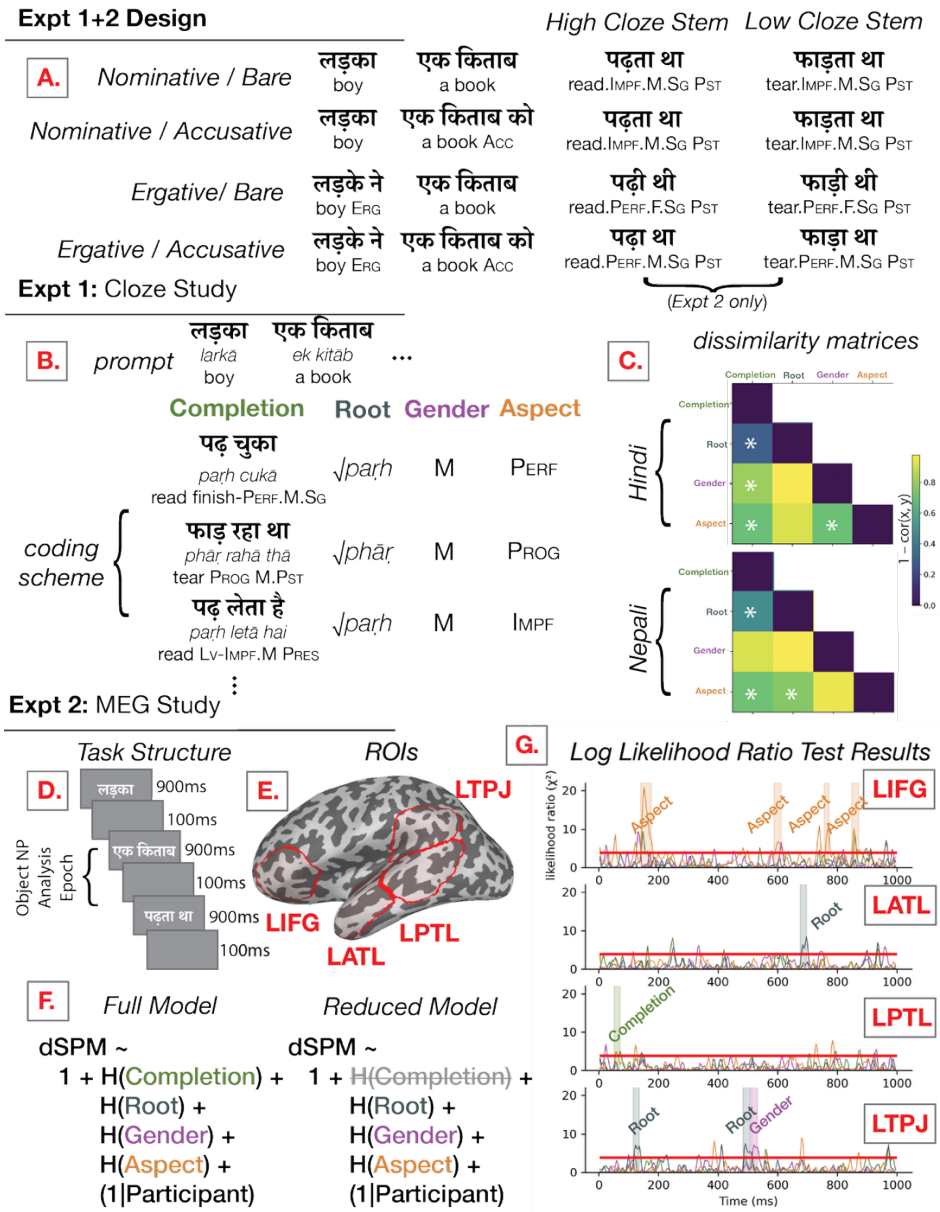

Fig 1. [A] Stimulus design for Experiments 1+2. [B] Example responses from the Cloze task. [C] Dissimilarity of entropy values from Experiment 1. [D] Task structure for MEG experiment. [E] ROIs plotted on a participant's MRI scan. [F] Full model vs. one reduced model for log likelihood ratio test. [G]. Significant clusters by ROI for log likelihood ratio test. Shaded areas indicate span of significant clusters at Bonferroni-corrected α = 0.0125.

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